

SHENGYANG TAOXIAN, China

WMO: 543424

Lat: 41.640N

Lon: 123.483E

Elev: 198

StdP: 14.59

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-6.1	-20.2	1.8	-5.2	-16.5	2.3	-1.5	20.8	25.8	18.3	24.5	3.9	140	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	89.7	73.9	87.7	73.6	85.7	72.4	79.0	85.5	77.5	83.8	75.9	81.9	10.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	142.0	83.9	75.3	133.8	81.8	73.7	126.3	79.9	42.8	85.8	41.0	83.4	39.5	81.2	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.3	18.9	16.4	DB	-16.1	93.6	6.0	2.8	-20.4	95.6	-23.9	97.2	-27.2	98.8	-31.6	100.8	(4)
			WB	-16.5	81.2	5.9	1.7	-20.7	82.5	-24.2	83.4	-27.5	84.4	-31.7	85.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	11.1	20.3	34.8	51.1	63.9	72.1	77.0	75.0	64.9	50.2	32.5	16.5	(6)
		DBStd	24.16	9.41	10.30	9.13	8.20	6.31	4.67	3.63	4.50	5.90	8.44	10.73	10.17	(7)
		HDD50	4270	1205	831	475	84	1	0	0	1	103	531	1039		(8)
		HDD65	7390	1670	1251	936	420	96	5	0	2	72	460	974	1504	(9)
		CDD50	3395	0	0	5	118	432	662	838	774	449	110	7	0	(10)
		CDD65	1040	0	0	0	4	63	217	373	311	70	2	0	0	(11)
		CDH74	8491	0	0	0	61	663	1732	3051	2391	570	23	0	0	(12)
		CDH80	2552	0	0	0	6	158	553	1011	733	90	1	0	0	(13)
		WSAvg	7.3	6.1	7.2	8.6	9.9	9.1	7.3	6.2	5.9	6.3	7.4	7.7	6.4	(14)
		PrecAvg	26.2	0.2	0.2	0.6	1.3	1.9	4.0	7.2	5.8	2.4	1.4	0.8	0.4	(15)
(16)	Precipitation	PrecMax	36.1	1.1	1.0	1.4	2.7	4.4	6.5	12.9	14.3	5.5	4.2	1.9	0.9	(16)
(17)		PrecMin	19.3	0.0	0.0	0.0	0.2	0.6	1.6	3.3	1.5	0.1	0.2	0.2	0.1	(17)
(18)		PrecStd	4.9	0.2	0.3	0.4	0.7	1.1	1.3	2.9	3.2	1.2	1.0	0.5	0.2	(18)
(19)																
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	39.4	50.2	65.8	80.4	88.2	91.8	93.3	91.7	85.8	77.0	64.4	46.1		(19)
(21)		0.4% MCWB	32.7	38.5	50.0	58.0	63.6	68.9	76.9	76.9	71.2	61.7	52.7	40.2		(20)
(22)		2% DB	35.4	44.5	58.8	75.0	84.2	88.2	89.9	88.1	82.5	73.1	57.6	40.7		(21)
(23)		2% MCWB	29.6	36.1	44.8	55.2	62.4	67.9	76.7	76.2	67.5	59.0	48.3	35.1		(22)
(24)		5% DB	31.7	40.8	53.8	70.2	80.8	85.9	87.7	86.2	80.4	68.3	53.7	37.0		(23)
(25)		5% MCWB	27.1	33.7	42.1	52.6	60.6	67.7	75.9	75.0	65.7	56.3	45.8	32.4		(24)
(26)		10% DB	28.0	37.2	49.9	66.4	77.2	82.7	85.7	84.2	77.2	64.6	49.7	33.5		(25)
(27)		10% MCWB	24.0	31.2	39.4	50.3	58.7	67.4	75.0	73.8	64.1	53.9	43.3	29.3		(26)
(28)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	34.2	40.6	50.9	61.1	68.4	74.5	81.6	81.0	74.3	64.0	53.9	40.7		(27)
(29)		0.4% MCDB	37.6	46.6	63.3	74.2	83.5	85.2	89.1	87.2	82.1	73.2	62.2	45.1		(28)
(30)		2% WB	29.9	36.8	46.4	57.2	65.3	72.5	79.5	79.0	71.1	60.7	50.1	35.5		(29)
(31)		2% MCDB	34.2	42.9	55.6	70.8	78.3	82.3	86.8	84.9	77.9	69.2	56.4	39.1		(30)
(32)		5% WB	27.0	34.0	43.1	54.5	63.2	71.0	78.0	77.7	69.0	58.4	47.0	32.2		(31)
(33)		5% MCDB	31.2	39.6	52.6	67.8	75.9	80.3	84.7	83.6	76.0	66.4	52.8	35.9		(32)
(34)		10% WB	23.7	31.2	40.1	51.8	61.1	69.7	76.4	76.2	66.8	55.8	43.0	29.1		(33)
(35)		10% MCDB	27.4	36.5	48.5	63.7	73.6	79.0	82.7	81.8	73.9	63.0	48.5	32.7		(34)
(36)	Mean Daily Temperature Range	MDBR	21.7	21.5	19.6	22.0	21.4	18.1	15.2	16.2	21.6	21.6	18.3	19.7		(35)
(37)		5% DB MCDBR	23.6	22.9	25.0	27.5	26.1	23.2	18.7	18.3	23.9	24.4	22.4	20.4		(36)
(38)		5% WB MCWBR	19.5	16.2	15.1	14.1	12.0	9.7	8.6	8.9	11.4	13.5	15.6	16.8		(37)
(39)		5% WB MCWBR	22.9	21.1	23.0	24.0	21.8	17.9	15.5	14.8	19.5	21.2	20.2	19.3		(38)
(40)	Clear-Sky Solar Irradiance	taub	0.374	0.441	0.499	0.600	0.650	0.727	0.706	0.618	0.541	0.551	0.465	0.396		(40)
(41)		taud	2.014	1.864	1.802	1.650	1.605	1.529	1.606	1.766	1.867	1.761	1.892	1.994		(41)
(42)		Ebn at Noon	229	230	233	219	213	197	200	213	219	192	193	207		(42)
(43)		Edh at Noon	37	51	61	76	82	89	81	67	57	56	43	36		(43)
(44)	All-Sky Solar Radiation	RadAvg	747	1020	1365	1615	1835	1674	1528	1498	1396	1039	717	607		(44)
(45)		RadStd	48	67	88	103	141	167	123	121	82	67	70	44		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page